

Final Project Report - Security for Networked Systems

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github link: <https://github.com/nishasarwar/network-security-project.git>

1 Executive Summary

Organizations often invest either too little or too much in cybersecurity due to unclear risk levels and lack of structured decision making. This project proposes a tiered security model that classifies organizations into Low, Mid, High security tiers based on size, privacy requirements, threats, budget, and technical expertise.

The report includes a complete decision matrix, recommended defenses for each tier, and three practical technical demonstrations:

1. A phishing simulation
2. A vulnerability scan using nmap
3. A firewall scenario using UFW

For demonstration purposes, all these simulations are done on Jetstream, server and client VM provided in the course.

2 Introduction

Cybersecurity threats continue to grow in complexity and frequency, still many organizations struggle to determine what level of defense is appropriate for them. Small organizations may overspend on complex systems they do not need, and large organizations may underestimate the threat exposures.

This project addresses that challenge by building a structured consultancy-styled framework to help organizations self assess their security needs and adopt appropriate defense level. The project includes hands-on demonstrations using Jetstream VMs running Ubuntu 22.04. These demonstrations show how different security tiers respond to real-world attacks.

3 Tiered Security Framework

The model uses three tiers:

Low Security Tier	Mid Security Tier	High Security Tier
• Small teams (less than 50 people) • Mostly public or non-sensitive data • Limited budget for security • Minimal threat exposure • No dedicated security staff	• Small Medium-scale organizations (50–250 people) • Mix of internal and customer data • Subject to moderate risk and regulations • At least one person with security responsibilities	• Large enterprises or high-value targets • Handle extremely sensitive, regulated, or financial data • Frequently targeted • Dedicated security teams and mature processes

Table 1: Tiered Security Framework: Organizational Characteristics

4 Threat Landscape by Tier

Low Tier Threats	Mid Tier Threats	High Tier Threats
• Phishing • Credential stuffing • Basic malware • Misconfiguration exploits	• Insider threats • Ransomware • Lateral Movement • Web application exploits	• Advanced Persistent Threats (APTs) • Zero-day vulnerabilities • Supply chain compromises • Sophisticated spear-phishing

Table 2: Threat Landscape by Security Tier

5 Mitigation Strategies by Tier

Low Tier Defenses	Mid Tier Defenses	High Tier Defenses
• Multi-factor authentication (MFA) • Basic firewall • Regulated system updates • Cloud managed security solutions • Security awareness training	• Network segmentation • SIEM or log monitoring • Scheduled vulnerability scanning • Role based access control • Incident response plan	• Zero trust architecture • 24/7 SOC monitoring • Threat intelligence feeds • Advanced EDR/XDR (Endpoint/Extended Detection and Response) • Red-teaming (simulation of real-world attacks by a group of security experts, the "red team") • periodic penetration testing

Table 3: Mitigation Strategies by Security Tier

6 Decision Matrix for Tier Classification

Organizations can score themselves in five dimensions:

Factor	Score 1 (Low)	Score 2 (Medium)	Score 3 (High)
Organizational Size	<50 employees	50–250	>250
Data Sensitivity	Mostly public	Some personal/financial	Highly sensitive/regulated
Threat Exposure	Rarely targeted	Occasional	Frequent / high-value target
Security Budget	Minimal	Moderate	Dedicated budget
Technical Expertise	No security staff	One IT/security staff	Security team

Table 4: Decision Matrix for Tier Classification

Scoring:

- 5-7 → Low Tier
- 8-11 → Mid Tier
- 12-15 → High Tier

7 Technical Demonstrations

All three demonstrations were conducted using Ubuntu 22.04.5 LTS on Jetstream. Commands and outputs are included so the experiment can be reproduced.

7.1 Demonstration A: Phishing Simulation

Objective: Show how a social engineering attack succeeds or fails depending on the organization's security level.

Phishing Email (Safe Example)

Subject: Important: Account Verification Needed

Dear User, Your account has been flagged for unusual login activity. Please verify your identity within 24 hours to avoid suspension. Verify Now: <http://secure-update-support.com/login>

Thank you, Support Team

Red Flags Identified

- Urgent tone
- Suspicious domain name
- Generic greeting
- Threat of account suspension

Tier Responses

Tier	Expected Response
Low Tier	Employees likely click the link; no phishing training; no email filtering.
Mid Tier	Basic filtering may flag it; some users recognize suspicious link.
High Tier	Advanced filtering blocks it; employees trained; domain reputation checks stop delivery.

Table 5: Expected Organizational Response to Phishing Attempt by Tier

Phishing attack flow

Attacker → Fake Email → User Inbox → User Clicks Or Reports → Security Response

7.2 Demonstration B: Vulnerability Scan (Nmap)

Objective: Show how a simple network scan reveals exposed services.

Commands Executed

- From Client VM: `nmap -sV -O 192.168.122.183`
- More Aggressive scan: `nmap -A 192.168.122.183`

Output

```

cadmin@i520-client:~$ sudo nmap -sV -O 192.168.122.183
Starting Nmap 7.80 ( https://nmap.org ) at 2025-12-05 21:36 EST
Nmap scan report for 192.168.122.183
Host is up (0.00086s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.52 ((Ubuntu))
MAC Address: 52:54:00:B7:18:7F (QEMU virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.80%E=4%D=12/5%OT=22%CT=1%CU=39723%PV=Y%D=1%DC=D%G=Y%M=525400%T
OS:M=693396CC%P=x86_64-pc-linux-gnu)SEQ(SP=105%GCD=1%ISR=106%TI=Z%CI=Z%II=I
OS:%TS=A)OPS(O1=MSB4ST11NW7%O2=MSB4ST11NW7%O3=MSB4NNT11NW7%O4=MSB4ST11NW7%O
OS:5=MSB4ST11NW7%O6=MSB4ST11)WIN(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6
OS:=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=MSB4NNSNN7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0
OS:%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=
OS:0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%
OS:S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(C
OS:R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=6%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=
OS:N%T=40%CD=S)

Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.26 seconds

```

Figure 1: Initial Nmap Scan Output

```

cadmin@i520-client:~$ nmap -A 192.168.122.183
Starting Nmap 7.80 ( https://nmap.org ) at 2025-12-05 21:38 EST
Nmap scan report for 192.168.122.183
Host is up (0.0044s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.52 ((Ubuntu))
|_http-server-header: Apache/2.4.52 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.30 seconds

```

Figure 2: Aggressive Nmap Scan Output

Interpretation: A vulnerability scan using Nmap was conducted on the server VM (192.168.122.183). The scan revealed two open services: SSH on port 22 and HTTP on port 80. SSH is exposed and could be targeted by brute-force attacks if not properly secured. The HTTP service is running the default Apache page, which is unnecessary for production and could expose information to attackers. Mitigation strategies include enabling key-based authentication for SSH, restricting access via firewall rules, keeping services updated, and replacing default web pages with the actual application content.

Tier Responses

Tier	Response
Low	No action; unaware of open ports.
Mid	Identify unnecessary services; schedule remediation.
High	Continuous scanning + automated alerts; remove RPCbind; enforce hardened SSH policies.

Table 6: Organizational Response to Vulnerability Scan by Security Tier

7.3 Demonstration C: Firewall Hardening (UFW)

Objective: Show the impact of enabling a firewall on reducing attack surface.

Step 1: Initial Scan (Before Firewall)

- nmap 192.168.122.183 (Shows several open ports)

```

cadmin@i520-client:~$ sudo nmap -sV -O 192.168.122.183
Starting Nmap 7.80 ( https://nmap.org ) at 2025-12-05 21:36 EST
Nmap scan report for 192.168.122.183
Host is up (0.00086s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http      Apache httpd 2.4.52 ((Ubuntu))
MAC Address: 52:54:00:B7:18:7F (QEMU virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.80%E=4%D=12/5%OT=22%CT=1%CU=39723%PV=Y%DS=1%DC=D%G=Y%M=525400%T
OS:M=693396CC%P=x86_64-pc-linux-gnu)SEQ(SP=105%GCD=1%ISR=106%TI=Z%CI=Z%II=I
OS:%TS=A)OPS(O1=M5B4ST11NW7%O2=M5B4ST11NW7%O3=M5B4NNT11NW7%O4=M5B4ST11NW7%O
OS:5=M5B4ST11NW7%O6=M5B4ST11)WIN(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6
OS:=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=MSB4NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0
OS:%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%Z=F%R%O=%RD=
OS:0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%
OS:S=A%Z=F%R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(
OS:R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=6%RID=6%RIPCK=6%RUCK=6%RUD=G)IE(R=Y%DFI=
OS:N%T=40%CD=S)

Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.26 seconds

```

Figure 3: Initial Nmap Scan Output Before Firewall

Step 2: Enable Firewall

```

sudo ufw enable
sudo ufw deny 22
sudo ufw allow 80
sudo ufw status

```

```

cadmin@i520-client:~$ sudo nmap -sV -O 192.168.122.183
Starting Nmap 7.80 ( https://nmap.org ) at 2025-12-05 21:46 EST
Nmap scan report for 192.168.122.183
Host is up (0.00078s latency).
Not shown: 999 filtered ports
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.4.52 ((Ubuntu))
MAC Address: 52:54:00:B7:18:7F (QEMU virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 2.6.32 (94%), Linux 3.10 - 4.11 (94%), Linux 3.2 - 4.9 (94%), Linux 3.4 - 3.10 (94%), Linux 2.6.32 - 3.10 (93%), Linux 2.6.32 - 3.13 (93%), Linux 3.10 (93%), Synology DiskStation Manager 5.2-5644 (92%), Linux 2.6.22 - 2.6.36 (91%), Linux 2.6.39 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.02 seconds

```

Figure 4: Nmap Scan After Firewall Hardening

Interpretation: After enabling the firewall (UFW) and blocking unnecessary ports, a follow-up Nmap scan shows that only port 80 (HTTP) is open, and all other previously exposed ports, such as SSH and RPCbind, are filtered. This demonstrates that firewall hardening effectively reduces the system’s attack surface. The OS detection results are less reliable due to only one open port, but the scan confirms that network-level access is restricted according to security best practices.

Tier Responses

Tier	Response
Low	No firewall or misconfigured; all ports exposed.
Mid	Basic UFW rules applied manually; some ports blocked.
High	Only required ports open (e.g., HTTP); all others filtered; continuous monitoring.

Table 7: Organizational Response to Firewall Hardening by Security Tier

8 Business Value Analysis

The tiered security framework provides several practical benefits for organizations:

Cost Efficiency Helps organizations spend on security where it's needed and avoid unnecessary tools.

Risk Reduction Aligns defenses with the level of threat, lowering the chance and impact of incidents.

Decision Clarity Provides a clear, structured way for executives to assess their organization's security posture.

Operational Improvement Demonstrates concrete steps organizations can take, even with limited resources, to improve security.

9 Conclusion

Cybersecurity is not one-size-fits-all. This project demonstrates a practical approach through a tiered framework, decision matrices, and real technical demonstrations using Jetstream VMs. The combination of social engineering, scanning, and firewall hardening effectively illustrates how different organizations should respond based on maturity and resources.

The framework can be expanded in the future by integrating automation, real-time threat intelligence, and machine-learning-based risk scoring.